

aivancity

School of AI & Data for Business & Society

Paris Region · Nice Côte d'Azur

Collaboration between aivancity and
Breda University of Applied Science
Presentation of aivancity and 1+1 MSc
Program

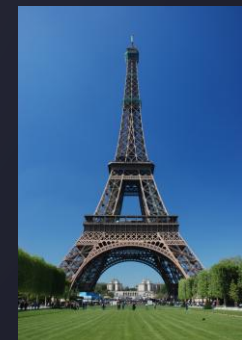
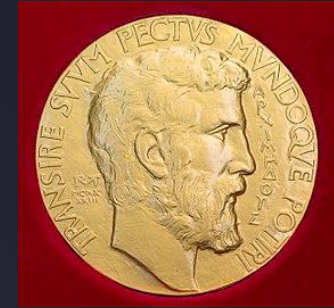


Why coming to France to Study?



Why France ?

- One of the world's top Economic & Education countries :
 - #6 GDP in the world and #2 in Europe,
 - #1 destination in Europe for international investment for the past 5 year
 - # 3 in Education in the world according to the Shanghai ranking
- A leading nation for research and innovation :
 - Ranked #1 in the world for fields medals (mathematics)
 - Ranked #4 for nobel price
- Recognized for the quality of its engineers :
 - 35 000 French engineers in the Silicon Valley
 - Several research centers of main Tech compaignies located in France (Meta, Google, Microsoft, IBM, TCS, etc.)
- An attractive destination with reasonably-priced education :
 - 430.000 students from the 5 continents
 - Several programs between 10K€ to 15 K€ per year with apprenticeship (fees paid by the companies)
- A great place to stay :
 - Its cultural life, its French cuisine, its fashion & luxury brands and Paris as the most visited city in the world



Artificial Intelligence is
changing the world



Definition of Big Data and Artificial Intelligence

- **Big data** is data that contains **greater Variety**, arriving in **increasing Volumes** and with **more Velocity**. **Value** and **Veracity** can be added to form the 5 Vs
- Artificial intelligence is a set of **algorithms** (mathematics) that gives a machine the **ability** to perceive, learn, organize memory and reason, allowing this machine **to perform tasks** usually dedicated to **humans**.



The 3 main domains of AI and Data

Data Science

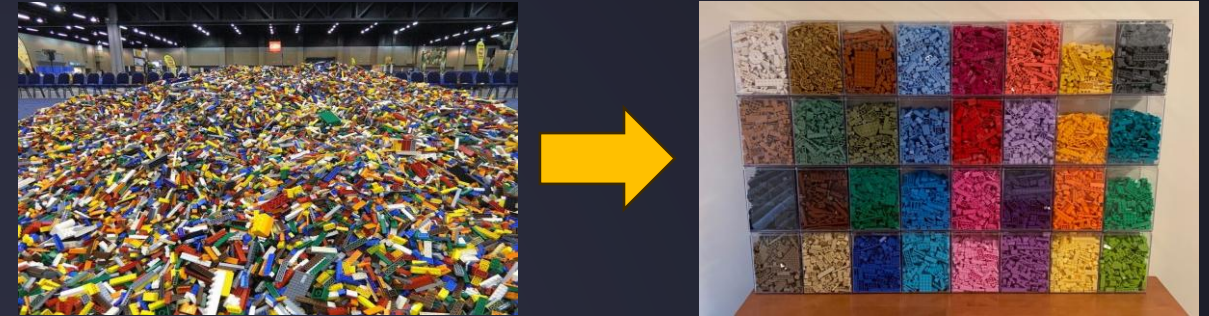
Data Engineering

Data Analytics

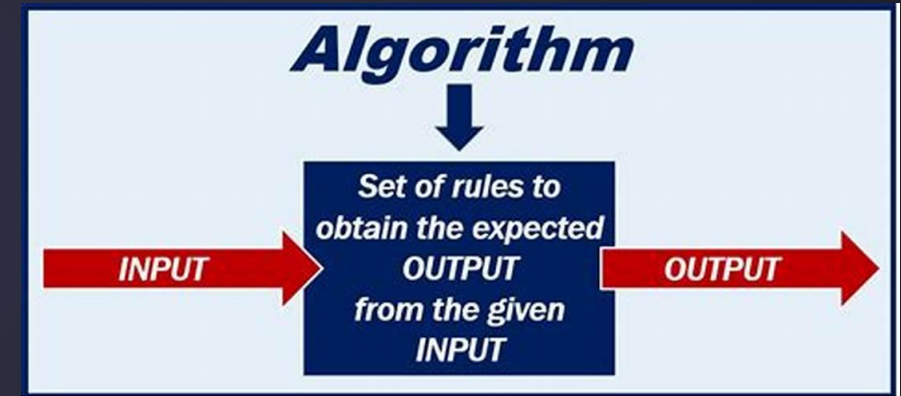
Data Scientist	Data Engineer	Data Analyst
		
uses statistics and machine learning to make predictions and answer key business questions	build and optimize the systems that allow data scientists and analysts to perform their work	deliver value by taking data, communicating the results to help make business decisions
Skills - Math, Programming, Statistics	Skills - Programming, BigData & Cloud	Skills - Communication, Business Knowledge
 	 	 
Tech - SQL, Python, R, Cloud	Tech - SQL, Python, Cloud, Distributed Computing	Tech - SQL, Excel, Tableau

The 3 main domains of AI and Data illustrated

Data Engineering : Convert unstructured and unordered data to structured and ordered data

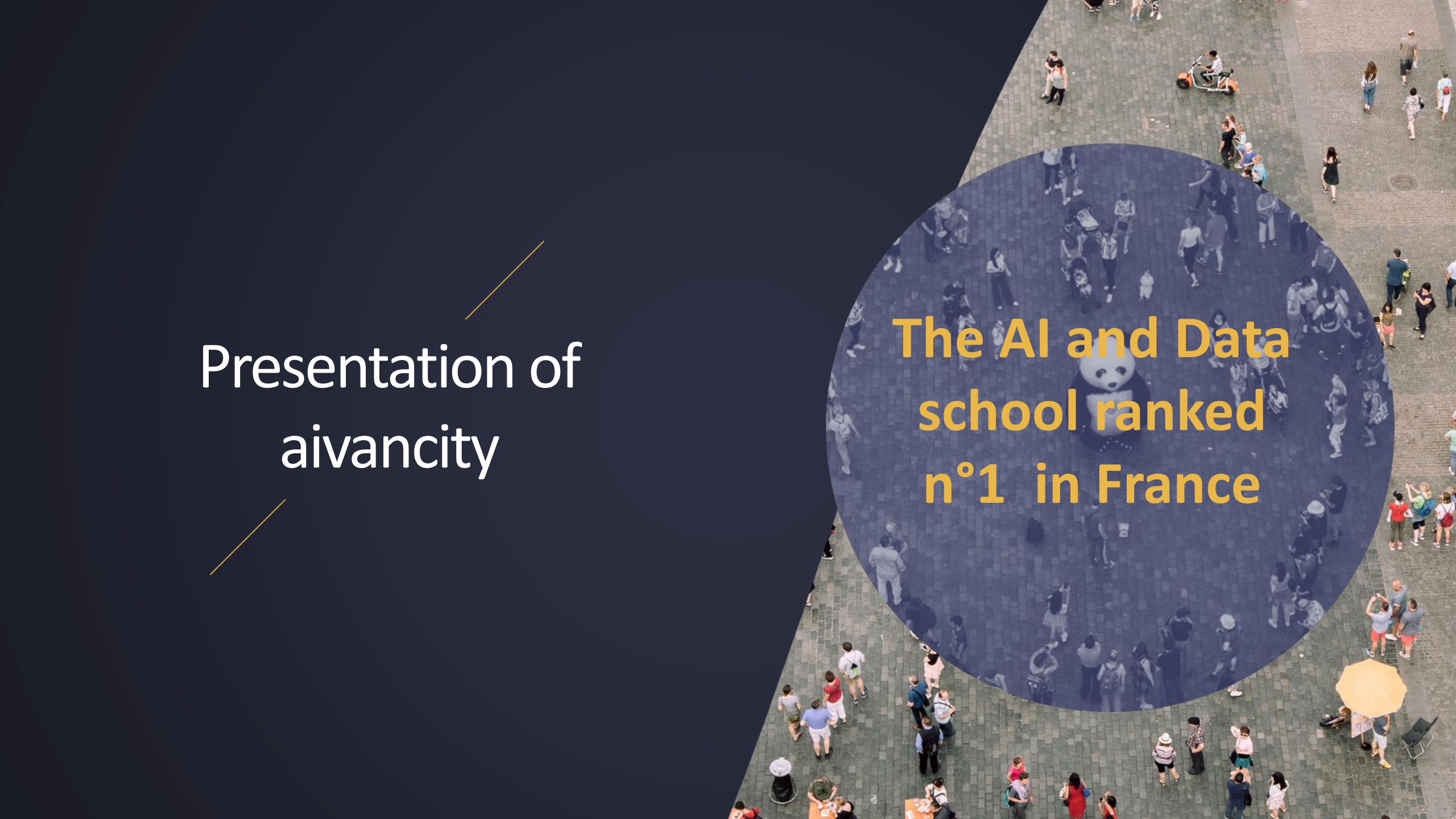


Data Science : Extract knowledge from massive datasets to make predictions (predictive AI) or generate new content (generative AI)



Data Analytics : Convert structured and ordered data to charts for decision making



An aerial photograph of a busy pedestrian plaza, likely in Paris, showing many people walking. A large, semi-transparent blue circle is overlaid on the right side of the image, containing text. The background is a dark blue gradient on the left side.

Presentation of
aivancity

**The AI and Data
school ranked
n°1 in France**

A specialized school with a different positioning

- aivancity has been created to meet the many challenges raised by the development of **Artificial Intelligence and Data**
- aivancity is a specialized French higher education Institution (“Grande Ecole”) whose mission is to prepare **students and professionals** to these new challenges
- aivancity is a **unique** school in France due to its **hybrid positioning: Technology / Business / Ethics**



A unique and innovative school



- The only school **100% dedicated to AI and Data** :
Data Science – Data Analytics – Data Engineering -
located in **Paris Region & Nice region**
- **A hands-on pedagogy** built under the tryptic **Technology-Business – Ethic** with a practical approach thanks to the **AI Clinic, Internships & Dual work studies**
- **An international school with 25+ nationalities**, students from **different background (economic, humanities, sciences)**, **more than 35% female** and no age limit (students up to 45yo)
- The possibility for every student to **come back to Campus and update skills and degree all life long** (Lifetime Update Guarantee Diploma®)
- The N°1 AI school in France with **all State recognitions**:
 - The **recognition** from the Ministry of Higher Education **of the school** and the **Visa for** its 5-year Grande Ecole Program
 - **3 RNCP recognitions** by the Ministry of Labour (Bachelor -> RNCP level 6 - Masters -> RNCP level 7)
- **Ranked n°1** AI School in France by EDUNIVERSAL in Jan 24
- Member of the **Campus France Forum** and **UGEI** (Union des Grandes Ecoles Indépendantes)
- A wide presence in France with **2 campuses in Paris Region and Nice**



Life on campus



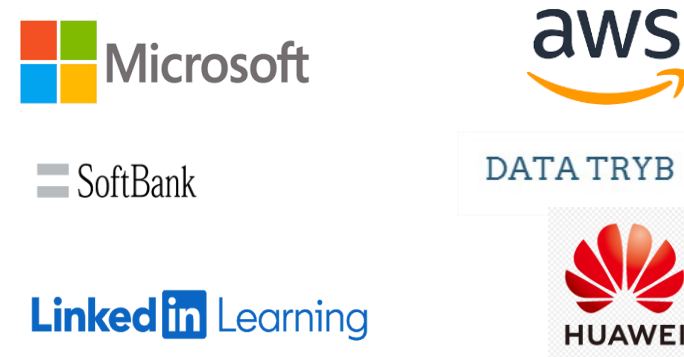
A few of our partners



Academic Partners



Technical collaborations



Institutional Partners

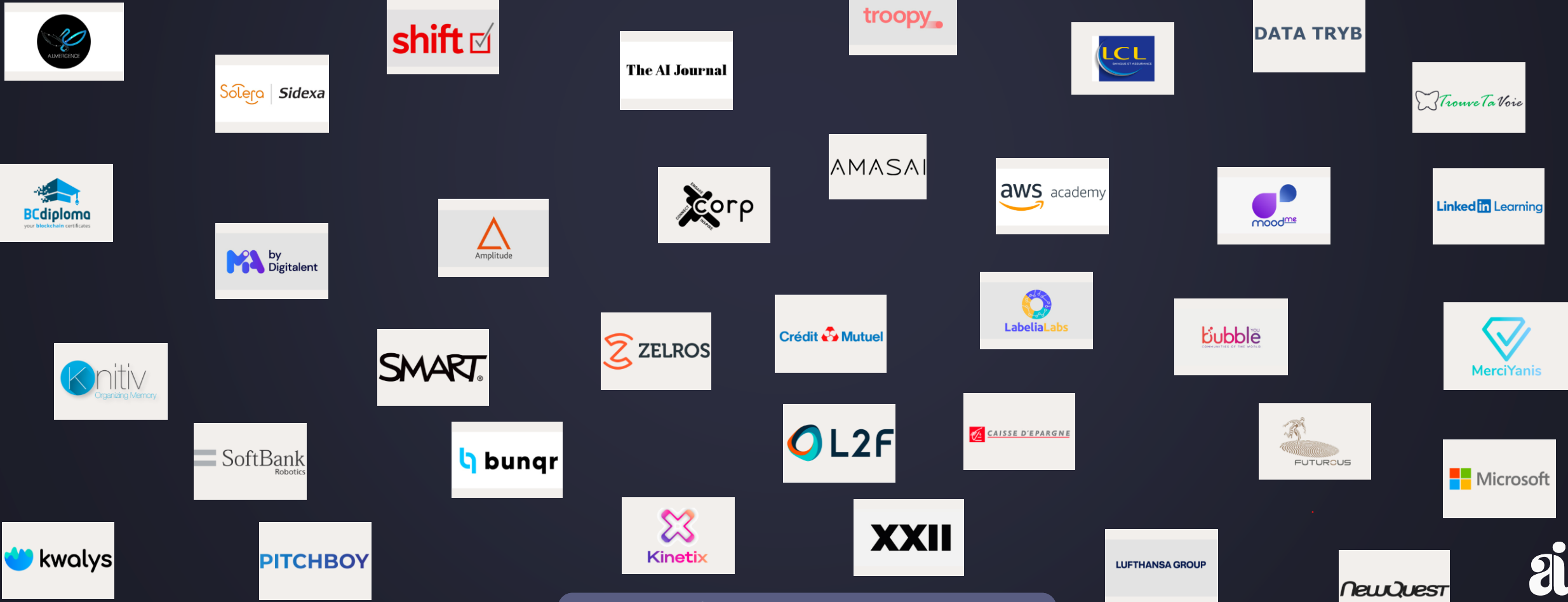


Banking partners



A few of our partners

Business Partners



aivancity different programs

*After high school,
after bachelor degree,
summer schools,
executive programs,
online, ...*



Grande Ecole Program

**Bachelor of Science or
Master's AI & Data Science**

3/5-year program

French or English

State Certif. – RNCP level 6 or 7

Master of Science

MSc Data engineering (2 years,
in english)

MSc Data Management (2
years in French or english)

**MSc AI for Business (part-time)
in French**

Executive Education

Short programs in 3 days and
certifying in 6 or 9 days

Tailor-made programs

MSc AI for Business (part-time)

Online programs

Free Mooc with paid
certification and Spocs

Certificates 100% On line

100% online degree programs

Summer programs

1 sessions July for High school
and University students

Immersive and certifying
programs

An aerial photograph of a busy city square, likely in London, showing a large crowd of people walking. A large, semi-transparent blue circle is overlaid on the right side of the image, containing a panda mascot. The background is a dark blue gradient on the left side.

aivancity
Programs for students

2025/2026
Outlook of our
programs

Programs for students at aivancity School of AI for Business & Society



Grande Ecole Program Master and Bachelor degree in AI & Data Science

- **Entry level:**
 - After High school or a Bachelor degree
- **Target :**
 - Students with an appetite for new tech. Background in Science or Humanities or Economics
- **Language :**
 - French or **English**
- **Campus**
 - Nice for French and **Paris for English**
- **Training duration :**
 - Full program in 5 years. Possible entry Y2/Y3/Y4
 - Apprenticeship for Y5 for every students and Y4 & Y5 for students that have started the program in Y1
- **Learning trip**
 - Y3 : In Morocco or India
 - Y4 : UC Berkeley (USA – Silicon Valley)
- **Degree awarded:**
 - BSc AI Developer (RNCP level 6 – HSD +3)
 - Master AI & Data Science (RNCP level 7 - HDS +5)
- **Academic Intake**
 - Main intake : Sept. (Possible late arrival early Nov.)
 - Late Intake : Jan. for PGE1
- **Fees**
 - From 10 500€ to 14 900€ per year



Master of Science Data Management (Data Analytics)

- **Entry level:**
 - After a Bachelor of Art (Business, Economics, commerce, etc.)
 - After Tech Bachelor (Computer science, application, etc.)
- **Target :**
 - Students with a business mind set and a strong appetite for new technologies
- **Language :**
 - French or **English**
- **Campus**
 - Nice or Paris for French and **Paris for English**
- **Training duration :**
 - 2 years apprenticeship for program in French
 - 2 years with 4/6 months internship for the program in English
- **Degree awarded:**
 - MSc Data Management (RNCP level 7 – HDS +5)
- **Academic intake:**
 - Jan.
- **Fees**
 - 12.500€ per year



Master of Science Data Engineering & Cloud Computing

- **Entry level:**
 - After Btech (4 years) or Bachelor of Science (3 years)
- **Target :**
 - Graduate students in science (computer science, mathematics, etc.)
- **Language :**
 - French or **English**
- **Campus**
 - Nice for French and **Paris for English**
- **Training duration :**
 - 2 years on a Dual work study basis
- **Learning trip (optional):**
 - 1 week in Europe.
- **Degree awarded:**
 - MSc Data Engineering (RNCP level 7 – HDS +5)
- **Academic intake:**
 - November ((Possible late arrival mid. Jan)
- **Fees**
 - 12.500€ per year



Presentation of the **Grande Ecole Program** (Integrated Master Program) at aivancity

YEAR 1

Data processing

🕒 500H of teaching + 120H of AI clinic

Computer Science

- Operating system management
- Introduction to databases
- Data analysis with Excel

IA tech & programmation

- Data acquisition, storage and analysis
- Data pre-processing and visualization
- Algorithms and programming in Python
- Machine Learning: use cases and applications

Math & Statistics for AI

- Statistical data analysis
- AI mathematics

Business management

- Business and market economics
- Corporate legal environment
- Introduction to corporate information systems

AI, Ethics & Society

- Engineering ethics
- AI Law
- AI philosophy

Langue et cultures

- English
- French as a foreign language

Projects & personal development

- AI Clinic
- Career coaching
- Internship 2 to 3 months

aivancity experience

- Associative, artistic and sporting activities
- Lecture series

YEAR 2

Data modeling

🕒 500H of teaching + 120H of AI clinic

Computer Science

- Servers and networks
- Information systems architect

AI tech & programming

- Machine learning
- Advanced programming with python
- Data visualization with R

Maths & statistics for AI

- Mathematical learning model
- Constrained optimization

Business management

- Marketing
- Accounting and financial information
- AI development project management

AI, Ethics & Society

- AI ethics
- Ethics by design
- Technology regulation

Language and culture

- English
- French as a foreign language

Projects & personal development

- AI Clinic
- Career coaching
- 3 to 4 month internship

aivancity experience

- Associative, artistic and sporting activities
- Lecture series
- Community Projects

YEAR 3

Data modeling

🕒 550H of teaching + 120H of AI clinic

Computer Science

- Cloud computing
- Information and data security
- Data Wrangling

IA tech & programmation

- Practical machine learning
- Coding for AI and Data Science
- DevOps for Artificial Intelligence

Mathematics and statistics for AI

- Mathematics of Deep Learning
- Time-Series Analysis

Business management

- Cost analysis & budgeting
- Business Game
- HR management & collective intelligence

AI, Ethics & Society

- Data governance
- Compliance and responsible AI
- Ethics of algorithms

Language and culture

- English
- Working in an international environment

Projects & personal development

- AI Clinic
- Career coaching
- 3 to 4 month internship
- Learning trip or Academic exchange

aivancity experience

- Associative, artistic and sporting activities
- Lecture series
- Community Projects

Presentation of the **Grande Ecole Program** (Integrated Master Program) at aivancity

YEAR 4

Deepen and develop

🕒 520H of teaching + 120H of AI clinic

Practical Deep Learning

- Machine learning Business Projects
- Deep learning: methods and applications
- Natural language processing
- Advanced data analysis and visualization

Deployed AI

- Big data in practice
- Deep learning: computer vision
- Deployment and maintenance of AI models

Business management

- Management of digital systems
- Supply chain management, smart operations
- Business strategy and digital transformation
- AI & Data science for Marketing
- Strategy & Business planning

AI, Ethics & society

- Data anonymization
- FairML
- Robustness
- XAI: Explainable Artificial Intelligence

Langues et cultures

- TOEIC preparation or French as a foreign language

Projects & personal development

- AI project Management
- AI Clinic
- Career coaching
- 3 to 4 month internship
- Learning trip: UC Berkeley

aivancity experience

- Associative, artistic and sporting activities
- Lecture series
- Community Projects

YEAR 5

Strategic vision and specialization

🕒 350H of teaching + 60H of AI clinic

Develop a responsible and strategic vision for AI

- Generative AI
- Cybersecurity and artificial intelligence
- Artificial Intelligence of Things
- Evolution of law in AI: practical cases
- Green Artificial Intelligence
- Artificial Intelligence and Human-Machine Interface
- Change management
- Data-driven business models
- The Future of Artificial Intelligence
- Structuring and management of an AI project

Applications of AI in different sectors

- AI-powered Customer Relationship Management
- Industrial AI
- AI in Agriculture
- AI for Health
- AI in Finance
- AI for Marketing

Projects & personal development

- Responsible leadership and management
- AI Clinic
- Career coaching
- Work-study report
- Final project or thesis
- Apprenticeship or internship 12 months

aivancity experience

- Lecture series

YEAR 1 (60 ECTS CREDITS)

Understanding data engineering - Managing a data project - Preparing for massive data

Certificate 1: Data analytics & programming

- Data Project Modeling and Design
- Advanced Relational DataBases
- Project Management: People and Technology

Certificate 2: Distributed Data System

- Distributed Algorithms
- Hadoop & Spark

Certificate 3: Data Computing

- Data analysis & visualization
- Cloud Computing
- Machine Learning

Certificate 4: Driving Data Strategy

- Web Security
- Data Laws & Regulations

YEAR 2 (60 ECTS CREDITS)

Mastering Cloud tools - Managing a massive data project - Understanding the Big Data ecosystem

Certificate 1: Big Data Management

- Java & Scala
- Data Pipeline (XML, DTD, XLS, JSON)
- Cloud computing

Certificate 2: Large Scale Data Services

- Hadoop & Spark
- Data mining
- Data Security

Certificate 3: Cloud Innovations

- Data analysis & visualization
- Big Data analytics
- Cryptography

Certificate 4: Cloud Data & Society

- Project Management: People and Technology
- Ethics of Data & Green Computing
- Data Regulation & Law

Presentation of the Data Management Program at aivancity

YEAR 1

DATA ANALYSIS	RESPONSIBLE DATA VIZUALISATION & REPORTING	DATA STRUCTURE & EXPLORATION	DATA FOR BUSINESS	STAKES	PROJECT & PERSONAL DEVELOPMENT
• Probability and statistics for data analysis • Data analysis in Python • Advanced Excel	• Data Visualisation with Power BI (preparation for Microsoft PL300 certification) • Data Reporting • Data law (CNIL, RGPD et DPO)	• Exploratory data analysis • Algorithmic Design & Modeling • Integration & Cloud	• Data project Management • Data driven decision making • Business Intelligence & Open Data	• Data law • Data ethics • Data security	• Career coaching • Data Clinic (autonomous) • Personalized learning

YEAR 2

ADVANCED DATA ANALYSIS	MACHINE LEARNING	ETHICS & SOCIETY	BUSINESS & ECONOMICS	PROJECT & PERSONAL DEVELOPMENT
• Data analysis and transformation • Causal Inference • Big Data & Cloud Computing • Web Data & Api	• Mathematics for ML • Foundations of Machine Learning & Metrics • ML Project Sprint	• AI ethics • Strategic data issues	• Data-Driven Business Models • Financial data analysis • Social Data & Social Listening	• Career coaching • Personalized learning • Graduation project

DATA MANAGEMENT FOR MARKETING TRACK	DATA MANAGEMENT FOR OPERATIONAL EXCELLENCE TRACK	CERTIFICATIONS	AIVANCITY EXPERIENCE
• Web Marketing and CRM • Web analytics • Social analytics • Scoring & predictive marketing	• Operations management • Supply chain analytics • Data for quality management • Data integration assessment: data warehouse, data lake and streaming data processing	• Preparation for Microsoft and AWS Professional certifications	• Career coaching • AI clinic • Graduation project

Presentation of the
collaboration program
between aivancity
and BUAS

2025/2026
Outlook of our
programs

Possible fields of collaboration between aivancity and JLU

- Summer schools, study and learning trips (exchange programs)
- Exchange of lecturers (from both Universities)
- Creation of a 1+1 Master program (In Artificial Intelligence or Data Engineering)
- *Promotion to other fields* (2-year MSc Data Management to students from Bachelor programs of other fields)
- *Future opportunities* (projects and research)

Presentation of the 1+1
MSc AI Engineering or
MSC Data Engineering
at aivancity and BUAS

2025/2026
Outlook of our
programs

Presentation of the 1+1 MSc AI Engineering OR Data Engineering program

GENERAL

- 1st year at BUAS and 2nd Year at aivancity
- Relationships with technology partners and companies, enabling students to work on real-life issues
- A diploma with a lifetime update guarantee© including 5 years free of charge
- Preparation for the best professional certifications in the field from Microsoft or Amazon

DEGREE FOR THE DATA ENGINEERING PROGRAM

- Master of Science in Data Engineering & Cloud Computing
- State-registered professional certification RNCP level 7 (bac+5/6 equivalent): «Expert infrastructure et traitement des données massives». registered under #37763 (on 19-07-2023).
- Professional certifications whose preparation is integrated into the program: One per year (AWS, Azure, Kafka, MongoDB, ...)

LEARNING GOALS FOR THE DATA ENGINEERING PROGRAM

- Add value to a team, company or organization through a combination of technical and managerial skills, knowledge of legal and ethical issues related to data engineering, and the ability to communicate effectively with professionals from different disciplines
- Master the Cloud ecosystem
- Identify patterns and gather insights from large and complex data sets
- Gain an in-depth understanding of Big Data foundations, methods and technologies
- Design and maintain a platform for processing large volumes of data
- Design a data warehouse and processing infrastructure
- Know how to use various analysis and programming tools: Python, R, Scala, Java, etc.
- Analyze, process and model massive data using statistical models to make them intelligible and exploitable.
- Manage a data engineering project
- Familiarize yourself with the laws and regulations that are relevant to data science and governance
- Integrate ethical and societal issues, such as questions of privacy, responsibility or human autonomy in decision-making

Presentation of the 1+1 MSc AI Engineering OR Data Engineering program

GENERAL

- 1st year at BUAS and 2nd year at aivancity
- Relationships with technology partners and companies, enabling students to work on real-life issues
- A diploma with a lifetime update guarantee© including 5 years free of charge
- Preparation for the best professional certifications in the field from Microsoft or Amazon

DEGREE FOR THE AI ENGINEERING PROGRAM

- Master of Science in AI Engineering
- State-registered professional certification RNCP level 7 (bac+5/6 equivalent): « Chef de projet en développement de solutions d'intelligence artificielle » registered under #38584 on 9-02-2024
- Professional certifications whose preparation is integrated into the program: One per year (AWS, Azure,, ...)

LEARNING GOALS FOR THE AI ENGINEERING PROGRAM

- Understand the company and its organization, and situate it in its environment
- Identify and define the challenges and impacts of the different areas of AI use
- Analyze and process data
- Implement an automated learning solution (machine and deep learning) :
- Lead the IT development of an AI project
- Enhance the value of AI project results and manage an artificial intelligence and/or data science project
- Take into account the multiple impacts and challenges of AI on the environment, society and the individual,
- Develop a responsible approach (legal issues relating to data, bias, transparency, robustness and explicability)
- Take a global approach to business issues,

Presentation of the 1+1 MSc AI Engineering OR Data Engineering program

Year 1 in Local University		Hours	ECTS
UE 1 - AI TECH (select 3)		54	6
1	Python Programming	18	2
2	R Programming	18	2
3	Data visualization with R	18	2
	Data Acquisition, Storage & Analysis	18	2
	Time-Series Analysis	18	2
UE 2 - AI TECH		168	27
4	Machine Learning	24	4
5	Data Analysis	24	4
6	Information & Data Security	24	4
7	Big Data	24	4
8	Cloud Computing	24	4
9	Coding for AI & Data Science	24	3
10	DevOps	24	4
UE 5 - Mathematics & Statistics		72	12
11	Essential Mathematics for AI & Data Science	24	4
12	Statistics for AI & Data Science	24	4
13	Mathematical foundations for Deep Learning	24	4
UE 6 - Business and Society		90	15
14	Data Ethics	18	3
15	Data Regulations	18	3
16	Digital Transformation	18	3
17	Fundamentals of Marketing	18	3
18	Fundamentals of Finance	18	3
UE 7 - Language		24	0
	French course	24	Pass/Fail
TOTAL GENERAL		408	60

Presentation of the 1+1 MSc AI Engineering OR Data Engineering Program

Year 1 at aivancity - Artificial Intelligence Engineering		Hours	ECTS
UE 1 - AI TECH - PRACTICAL ML/DL		96	12
1	Practical Machine Learning	24	3
2	Natural Language Processing	24	3
3	Deep learning: methods and applications	24	3
4	Advanced Data Analysis and Visualization	24	3
UE 2 - AI TECH - DEPLOYED AI		72	9
5	Big Data in Practice	24	3
6	Deep learning: computer vision	24	3
7	Deployment and maintenance of artificial intelligence models	24	3
UE 3 : BUSINESS MANAGEMENT (2 courses, depending on the period)		36	4
8	Supply chain management & smart operations	18	2
9	Business strategy and digital transformation	18	2
	Management of digital systems	18	2
	AI & Data science for marketing	18	2
	Responsible leadership & management	18	2
	Strategy & Business planning	18	2
UE 4 - AI, ETHICS & SOCIETY		90	10
10	Artificial Intelligence: Laws and regulations	18	2
11	Robustness	18	2
12	FairML	18	2
13	XAI: Explainable Artificial Intelligence	18	2
14	Data anonymization	18	2
UE 5 - Develop a responsible and strategic vision for AI (2 courses, depending on the period)		36	4
15	Artificial intelligence of things	18	2
16	Green artificial intelligence	18	2
	Artificial Intelligence and Human Machine Interface	18	2
	Generative AI	18	2
	The Future of Artificial Intelligence	18	2
	Change Management	18	2
	Cyber security and artificial intelligence	18	2
	Data-driven Business Models	18	2
UE 6 - Projects and Personal Development		88	21
	Welcome Seminar	8	
	AI Clinic	30	10
	French course (FLE)	24	Pass/Fail
	End-of-term Project	26	11
TOTAL GENERAL		418	60

Year 2 at aivancity : Data Engineering & Cloud Computing		Hours	ECTS
CERTIFICATE 0 : Boarding aivancity		54	6
	Java, programming, design, modeling	18	2
	Data Driven Decision Making	18	2
	Business Intelligence and Data Management	12	2
	Back-to-school seminar	6	0
CERTIFICATE 1 : Implementing Big Data project engineering		96	12
1	Big Data in Practice	24	3
2	Cloud computing	24	3
3	Advanced Data Analysis & Visualization	24	3
4	Project Management: People and Technology	24	3
CERTIFICATE 2 : Collect, prepare and secure data		72	9
5	Data Security	24	3
6	Data mining	24	3
7	Data Pipeline (XML, DTD, XLS, JSON)	24	3
CERTIFICATE 3 : Process and analyze large volumes of data		72	9
8	Java & Scala	24	3
9	Hadoop & Spark	24	3
10	Cryptography	24	3
CERTIFICATE 4 : Master business, legal and ethical project issues		36	4
11	Ethics of Data & Green Computing	18	2
12	Data Regulation & Law	18	2
PERSONAL DEVELOPMENT		80	20
	LinkedIn Learning - The foundations of DevOps	3	
	LinkedIn Learning - NoSQL : Big Data Management	3	
	French course (FLE)	24	Pass/Fail
	AWS Certified Cloud Practitioner	10	Pass/Fail
	<u>DP-203: Data Engineering on Microsoft Azure</u>	10	Pass/Fail
	End-of-term Project	30	20
TOTAL GENERAL		410	60

aivancity

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Paris Region · Nice Côte d'Azur

Thank You !

Contact us

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<https://www.aivancity.ai/en>



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L'IA a son adresse

